SOFTWARE REQUIREMENTS SPECIFICATION

Master Client Baseline Agreement & Front-End Architecture Contract

IEEE 830 STANDARD

Project Name	*[Insert Web Application Title]*	**Platform / Domain**	Web Platform (Responsive Front-End)
Client / Sponsor	*[Client Organization / Name]*	**Student Lead**	*[Student Developer Name & ID]*
Academic Course	25CS208 – Web Technology	**Faculty Mentor**	*[Faculty Supervisor Name]*

1. EXECUTIVE SUMMARY & SCOPE BOUNDARIES

This Software Requirements Specification (SRS) establishes the baseline technical contract between the Student Developer and the Client. It outlines client-side functional behavior, responsive layouts, data integration schemas, quality criteria, and cost structures.

In-Scope Commitments (Front-End Specification)	Out-of-Scope (Backend / Infrastructure)
• Responsive HTML5/CSS3 layouts & component architecture. • Form validation via JS Regex & dynamic DOM feedback. • REST API integration (`fetch()`) & `localStorage` CRUD. • Optional 100% On-Device AI Chatbot (Transformers.js / WASM).	• Custom server-side database hosting & backend controllers. • Third-party payment gateway server processing. • Native mobile app compilation (iOS / Android). • Long-term server administration post-hand-off.

2. CROSS-FUNCTIONAL ENGINEERING ROLES & GOVERNANCE

Cost Management (Project Manager Role)

- Calculates labor costs: **Est. 72 Hours @ ₹800/hr = ₹57,600**.
- Prevents scope creep; manages timeline milestone risks.
- Audits free-tier cloud limits (₹0/mo Vercel / Netlify hosting).

Revenue Management (Product Owner Role)

- Defines monetization strategy (SaaS, Freemium, Commission).
- Designs UI paywalls, pricing tables, & order summaries.
- Establishes commercial value proposition for client target market.

3. FRONT-END FUNCTIONAL REQUIREMENTS (FR)

FR-ID	Feature Module	Client-Side Logic & User Experience	Acceptance Criterion
FR-01	Responsive Navigation	Flexbox/Grid menu; toggle drawer on mobile (<768px); ARIA focus labels.	Passes mobile audit on all devices.
FR-02	Dynamic Input Validation	Client-side Regex validation on blur event; custom inline error messaging.	Prevents submit on bad input.
FR-03	REST API Data Rendering	Async `fetch()` requests; dynamic DOM creation; loading state indicators.	Data populates under 2.0s.
FR-04	On-Device AI Assistant	Client-side WASM inference (Transformers.js); offline chat session persistence.	Generates responses offline.

4. NON-FUNCTIONAL & QUALITY STANDARDS (NFR)

Performance	Initial load time < 2.0s. Google Lighthouse Performance score ≥ 90.
Accessibility (a11y)	WCAG 2.1 Level AA compliance. Keyboard focus navigation (`Tab`) & min 4.5:1 text contrast ratio.
Browser Support	Tested and verified visual consistency on Chrome (v110+), Safari (v16+), Firefox (v112+), Edge (v110+).

5. COST PROPOSAL & DELIVERY MILESTONES

Item / Phase	Scope / Deliverables	Schedule	Estimated Cost
One-Time Labor	Requirements, Wireframes, HTML5/CSS3, JS Interactivity, QA Audit (72 hrs)	Weeks 1 – 12	₹57,600
Domain & Infrastructure	Custom `.in` / `.com` Domain (₹1,200/yr) + Free Static Cloud Hosting (₹0/mo)	Annual Renewal	₹1,200 / yr

6. CLIENT SIGN-OFF & SCOPE AUTHORIZATION

By signing below, the Client Representative and Student Developer approve the requirements, scope boundaries, non-functional standards, and financial estimates outlined in this SRS document.

Client Representative Authorization:

Signature: _____
Name: *[Client Name]* | Date: _____

Student Lead Developer Acknowledgment:

Signature: _____
Name: *[Lead Student Name]* | Date: _____